

Fixed Forbidden Distances in Involutions

Uniform expansions, two Gaussian sectors,
and the adjacent-transposition law

Research article and reproducibility companion (source 19)

research note on OEIS A170941 and A217876 (source 26)

merged research report, OEIS A239144, A170941, A217876

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Abstract

The fixed columns of OEIS A239144 count involutions whose transpositions have distance greater than a fixed integer r . We give a self-contained adaptation of classical matching-polynomial and Gaussian-duality methods that produces their asymptotic expansions to every prescribed finite order. The underlying theorem is uniform over all forbidden graphs of fixed bounded degree, with coefficients evaluated at the graph's actual normalized matching-root moments. It supplies a complex probability-generating-function expansion and exponentially weighted signed-probability error bounds. For avoidance counts, two explicitly cutoff-defined positive Gaussian sectors have separately controlled expansions, with relative weights $e^{\sqrt{n}}$ and $e^{-\sqrt{n}}$. A connected-cluster argument proves exact eventual affine dependence of each fixed-distance local statistic, turning the general moving-statistic expansion into constant-coefficient column expansions. We derive first corrections and distinguish smooth, log-linear, and integer-threshold inverses. The exact Gaussian identity, root-moment technique, and high-order structural-correction methodology are established prior work; the present scope is an explicit involution-family refinement, without a worldwide-priority claim.

Merged material (source 26, Section 9). For forbidden distance one, let X_n count the adjacent transpositions in a uniformly chosen involution of $[n]$ (OEIS A217876; its zero column is A170941). Its probability generating function has an expansion to every fixed order in $n^{-1/2}$, uniformly on each compact subset of the complex plane, with rational polynomial coefficients obtained from Gaussian moments and a Touchard transform; Cauchy estimates give a remainder of the same order in every fixed exponentially weighted ℓ^1 norm of the probability sequence. The source computes the resulting total-variation corrections, the avoidance count, an all-finite-orders Lambert- W /Newton inverse with precise integer and log-linear conventions, and an exact Gaussian-Chebyshev decomposition that isolates a parity-alternating sector of relative size $e^{-2\sqrt{n}}$, with a separate Poincaré expansion for each exact sector. These are the case $G = P_n$ of the general theorems, re-proved there by a different route and refined. The avoidance enumeration goes back to Haslinger and Stadler; Ganjtabesh and Steyaert (2011) give exact inclusion-exclusion and the zero-count limit $A_{n,1}/I_n \rightarrow e^{-1}$; Li's 2026 preprint explicitly proves the full Poisson law; the first absolute correction 31/24 was recorded by Kotěšovec in OEIS in 2014. Source 26 develops refinements and error bounds, without claiming exhaustive historical priority or optimal truncation.

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Provenance, notation and status of this report

[Added 2 October 2026, batch 77.] This report was built in the write phase of batch 77 of ProveIt's incoming-reports intake (cluster P4) from **two manuscripts**, both delivered in the arrival commit 096ee7b87 and placed by commit d0e6008d9. Numbers are the batch-77 manuscript numbers.

No.	archive (main file)	pin	printed here as
19	<code>fixed-forbidden-distance-involutions.zip</code> (<code>article.tex</code> , 13-page PDF)	none named	the base: Sections 1–8, unchanged in substance
26	<code>involution-refinements-attribution-revision.zip</code> (<code>report.tex</code> , 15-page PDF)	none named	Section 9 and Appendix A; its credits in Section 1.1
27	<code>involution-refinements-reproducibility.zip</code> (<code>report.tex</code> , 15-page PDF)	none named	not printed: superseded by 26

Neither manuscript names a ProveIt commit or continues a repository path; both are dated 2 October 2026 and name neither an author nor a tool (author lines as above). Manuscript 27 is the earlier edition of 26: 26’s revision record (`data/26-adjacent-revision-integrity.json`) holds the SHA-256 of 27’s `report.tex`, and the two differ only in attribution (26 adds Ganjtabesh–Steyaert 2011 for the zero-count limit); nothing of 27 is shipped.

Why one report, and where the merge chose. Source 26 proves, for the path P_n (forbidden distance $r = 1$), the theorems that source 19 proves for every forbidden graph of bounded degree, with the same coefficients 31/24, $-55/48$, 1717/1920. Source 19 is the base, being the more general. Every result, proof, example, remark, limitation and question of source 26 that is not literally a special case of a statement of source 19 with the same proof is printed in Section 9: the full law of X_n with the polynomials $Q_j(\ell)$, the total-variation corollary, the exact moments and triangle formula, a second proof of the law (by growing-moment summation rather than matching roots), the absolute and logarithmic avoidance coefficients, a second exact sector decomposition (Chebyshev, with a sharper residual) and the sector ratio, the log-linear inverse with its explicit chord term, the source’s checks, table and questions, and its coefficients through order eight. Source 26’s smooth root, Newton algorithm and two-ceiling enclosure coincide with source 19’s at $r = 1$ (same statements, same proofs) and are printed once, in Section 6. The historical credits of both sources are stated once, in Section 1.1.

Status. AI-assisted (presumed: the deliveries name no author or tool), unrefereed, and not formalized in any proof assistant. Placement in ProveIt gives it no formal status. Source 19’s `19-distance-review.md` reviews source 19’s article source (by its delivered hash), not this merged text.

Notation. Source 19’s notation is used throughout. The merged material of source 26 reuses several of its letters with other meanings; the convention below fixes them. Symbols renamed from source 26 are marked $*$.

symbol here	meaning	do not read as
$\varepsilon = n^{-1/2}$	both sources (t in source 26*)	—
$T(n, r)$	the A239144 triangle, $T(n, r) = A_{n,r}$	$T^{\text{adj}}(n, \ell)$
$T^{\text{adj}}(n, \ell)^*$	the A217876 triangle, $I_n \mathbb{P}(X_n = \ell)$ (source 26's $T(n, \ell)$)	$T(n, r)$; here $T^{\text{adj}}(n, 0) = A_{n,1} = a_n$
$P_h(w; a)$	source 19's universal polynomials, after extracting $e^{a_1(G)w}$, $a_1(P_n) = 1 - 1/n$	$P_j(w)$
$P_j(w), Q_j(\ell)$	source 26's constant-coefficient polynomials at $r = 1$, after extracting e^w (as in (26))	$P_j(w; a(P_n))$, which moves with n
$K_G^\sigma, C_G, L_n^\sigma$	source 19's exterior sectors (cutoff B), central term, scale $\propto e^{-1/4-a_1}$	the Chebyshev sectors
$K_n^{\sigma*}, \tau_n^{\sigma*}, \Xi_n^*, \mathcal{L}_n^\sigma$	source 26's Chebyshev sectors K_n^σ , tails T_n^σ , residual R_n ; scale $\propto e^{-5/4}$	$K_{P_n}^\sigma = K_n^\sigma - \tau_n^\sigma$; $L_n^\sigma = e^{1/n} \mathcal{L}_n^\sigma$ at $r = 1$
$D_h(a);$ $A_r = D_r(0)$	sector polynomials; unrestricted-involution coefficients (both sources)	$\hat{D}_{m,\sigma}^*$, source 26's $D_{r,\sigma}$, equal to d_m
d_m, β_j	source 26's absolute and logarithmic avoidance coefficients	$\beta_j(S)$ of Theorem 5.1; but $\beta_j = \gamma_{1,j}$
$\kappa_j^* = P_j(-1)$	source 26's r_j	the distance r
m, i	summation or iteration indices of source 26* (there r)	the distance r , which is 1 in Section 9
$\mathcal{U}_n^*$	Chebyshev polynomial of the second kind (source 26's U_n)	$U \sim N(1/2, 1/2)$, which both sources use
$\mathcal{H}^*, \mathcal{F}^*, \mathcal{F}_J^*, \mathcal{F}_\sigma^*$	source 26's series H, F, F_J, F_σ	$H_r(w; a), F_{G,w}(y)$ of source 19
$h_n^*, g_\sigma^*, y_{n,\sigma}^*$	source 26's $\phi_n, a_\sigma, r_{n,\sigma}$	the density ϕ , the statistics a_j , the distance r
$\hat{p}_{h,\sigma}^*, \hat{b}_{m,\sigma}^*$	source 26's sector generators $p_{h,\sigma}, b_{r,\sigma}$	$p_h(u; a), b_h(u; a)$; at $r = 1$ they differ by the moving a_j
R	radius of the compact w -set (source 26's B^*)	the root bound B of (4)
$\mathcal{D}_J^*; c_J, c'_J, \Omega_J,$ ϑ_k, k_0, ℓ_0	source 26's $D_J(\ell, t)$; local constants $B_J, D_J, M_J, \rho_k, K, L^*$	D_h ; B, M_k , roots ρ_i , sectors K, L
$L_{1,J}, x_{1,J}, \tilde{x}_{1,J},$ $\mathcal{A}_1, N_1$	source 26's $L_J, x_J, \tilde{x}_J, \mathcal{A}, N$ (identical objects)	—

Typography: source 26's probability symbol is printed $\mathbb{P}$, its Gaussian density φ is printed ϕ , and its Pois is printed Poisson; “Kotesovec” in source 19 is printed Kotěšovec. No other symbol was renamed, and no statement, proof or number of either source was changed. All labels carry the prefix **fdi**: (source 19's material and the notes added here) or **fdi:adj**: (source 26's material).

1 The fixed distance problem and its precedents

An involution on $[n] = \{1, \dots, n\}$ consists of fixed points and disjoint transpositions. Write I_n for the number of all involutions, with $I_0 = I_1 = 1$ and

$$I_n = I_{n-1} + (n-1)I_{n-2}. \quad (1)$$

For a fixed nonnegative integer r , let $A_{n,r}$ count those involutions for which every transposition (i, j) satisfies $|i - j| > r$. Thus

$$A_{n,r} = T(n, r) \quad \text{in OEIS A239144,} \quad A_{n,0} = I_n, \quad A_{n,1} = \text{A170941}(n).$$

This is the existing Arndt–Heinz counting family recorded in 2014, not a newly defined sequence [1]. The statements below take $n \rightarrow \infty$ with r fixed. They make no assertion about the triangular diagonal or an increasing $r = r(n)$.

The first practical consequence of the analysis is

$$\frac{A_{n,r}}{I_n} = e^{-r} \left[1 + \frac{r}{\sqrt{n}} - \frac{r}{2n} - \frac{r(8r^2 - 12r + 3)}{24n^{3/2}} + O_r(n^{-2}) \right]. \quad (2)$$

Every higher coefficient is effectively computable. For a fixed finite set S of forbidden positive distances, rather than all distances $1, \dots, r$, the same all-orders conclusion holds, with different computable constants.

1.1 Historical boundary

The machinery used here has a substantial history. Heilmann and Lieb established the real-rootedness and root bounds for matching polynomials [7]. Godsil developed matching duality and the connection between matching roots and walks in rooted path trees [8, 9]. Lass’s exposition explicitly attributes the shifted Gaussian duality to Godsil’s 1981 Theorem 2.4 and states it in a form whose evaluation at 1 is exactly the all-matchings complement identity used here [10]. The shift by 1 is therefore already classical.

Godsil and McKay use bounded matching roots, walk power sums, and saddle integration to obtain structural asymptotic corrections [11]. Especially close prior work is McLeod’s 2007 thesis, Chapter 4, and its 2010 journal continuation [12, 13]. The thesis’s Theorem 4.17 gives explicit corrections through n^{-5} and an $O(k^6/n^6)$ remainder in the exponent for perfect matchings in the complement of a k -regular graph on $2n$ vertices, with $k = O(n^{1-\delta})$. Theorem 4.18 expresses structural terms through short subgraphs. These are substantial high-order local-statistic results, not merely leading equivalents. The corresponding theorem pages were checked in the primary thesis PDF. The final journal text has not been compared word for word with that precursor.

For adjacent avoidance, Ganjtabesh and Steyaert already give the selected-pair counting formula and prove $A_{n,1}/I_n \rightarrow e^{-1}$ [14]. Kotěšovec’s 2014 contribution to A170941 records the first absolute correction $31/(24\sqrt{n})$ [2]. Li’s 2026 manuscript proves convergence of the full adjacent-transposition count to Poisson(1) [15]. Neither the leading $1/e$ nor the first adjacent correction is asserted as new here.

[Added 2 October 2026, batch 77.] *Credits of source 26, stated once for the whole report.* The inspected 1997 author preprint of Haslinger and Stadler, Section 3.1, printed pages 16–17, gives the avoidance recurrence and growth discussion; the related paper appeared in 1999 [5, 6]. In the present indexing the recurrence is (35). Ganjtabesh and Steyaert’s 2011 paper [14, Section 5] already gives the selected-pair counting numerator in equation (17), avoidance inclusion–exclusion in equation (18), and the zero-count limit in Theorem 3 and equation (25), printed pp. 409–412. OEIS credits Václav Kotěšovec on 10 September 2014 with the absolute first correction $31/(24\sqrt{n})$ [2]. Li’s manuscript dated 29 August 2026 explicitly proves the full Poisson(1) law using factorial moments [15, theorem labelled `thm:poisson`]; the zero-count limit has the earlier 2011 source. The lingering conjectural wording on the OEIS pages is not evidence that either

leading limit remains open. Source 26’s targeted inspection did not locate the uniform all-orders probability theorem or the exact separate-sector refinement for the adjacent case; this is a statement about inspected sources, not a worldwide novelty certificate. In particular, the final typeset 1999 paper was not verified to be textually identical to the recovered author preprint, and the related Régnier and Nebel references were not exhaustively inspected (`data/26-adjacent-sources.json` records the URLs and the hashes of the inspected bytes, without redistributing third-party texts). Both sources credit Ganjtabesh–Steyaert for the zero-count limit and Li for the full Poisson law; this credit chain is the one of source 26’s attribution revision (see the provenance section). The finite Gaussian identity $I_m = \mathbb{E}(1 + Z)^m$ of Lemma 2.1 is classical; it is also stated in the repository’s *Combinatorial Transseries Inverses* volume, chapter `ct:ch:involutions`, equations `t2:eq:involution-enumeration` and `t2:eq:involution-A` [20], and no novelty is claimed for it here.

The useful scope of this article is the combined uniform, arbitrary-finite-order unrestricted-involution result: arbitrary bounded-degree forbidden graphs with moving local statistics, complex PGFs, weighted signed-law errors, and separately controlled unequal Gaussian sectors; together with fixed-distance stabilization and carefully specified inverses. A bounded inspection of the sources just named did not locate that combined theorem. This is a statement of inspected scope, not proof of global novelty. In particular, related perfect-matching Poisson limits [16], finite-order partial-matching estimates [17], and root-moment algorithms [18] are important precedents. No claim is made that this article invents the integral representation, root-moment method, or high-order structural-correction approach.

2 Exact identities and finite local statistics

Let G be a simple graph on $n \geq 1$ vertices and let $\Delta(G) \leq \Delta$, where Δ is fixed independently of n . The edges of G are forbidden transpositions. Let $M_k = M_k(G)$ be the number of k -edge matchings, with $M_0 = 1$, and define

$$\mu_G(y) = \sum_k (-1)^k M_k y^{n-2k}, \quad F_{G,w}(y) = \sum_k M_k w^k y^{n-2k}.$$

Let $X = X_G$ count edges of G occurring as transpositions in a uniformly chosen involution. The avoidance count is $A_G = I_n \mathbb{P}(X = 0)$.

Lemma 2.1 (Marking and Gaussian duality). *For every complex w ,*

$$I_n \mathbb{E}(1 + w)^X = \sum_k M_k w^k I_{n-2k} = \mathbb{E} F_{G,w}(1 + Z), \quad Z \sim N(0, 1). \quad (3)$$

In particular $A_G = \mathbb{E} \mu_G(1 + Z)$.

Proof. Expand $(1 + w)^X$ by selecting a subset of occurring forbidden edges. A selected k -matching has I_{n-2k} completions. Summing over the M_k selected matchings proves the first equality. The finite identity $I_m = \mathbb{E}(1 + Z)^m$ follows from Gaussian moments or the exponential generating function $\exp(t + t^2/2)$. This proves the second equality without an infinite-series interchange. The avoidance identity is the specialization $w = -1$ of the classical matching duality discussed above. $\square$

Write the n roots of μ_G , with multiplicity, as $\rho_1, \dots, \rho_n$. The Heilmann–Lieb bound allows the conservative common choice

$$B = 2\sqrt{\max(\Delta - 1, 1)}, \quad |\rho_i| \leq B. \quad (4)$$

The convention covers degrees 0 and 1 as well as the usual degree-at-least-two case. Define

$$a_j(G) = \frac{1}{2jn} \sum_{i=1}^n \rho_i^{2j}, \quad 0 \leq a_j(G) \leq \frac{B^{2j}}{2j} \quad (j \geq 1). \quad (5)$$

If ν is the maximum matching size, parity and real-rootedness give

$$\mu_G(y) = y^{n-2\nu} \prod_{i=1}^{\nu} (y^2 - \eta_i^2), \quad F_{G,w}(y) = y^{n-2\nu} \prod_{i=1}^{\nu} (y^2 + w\eta_i^2), \quad (6)$$

where the positive roots η_i are listed with multiplicity. Consequently

$$\log \mu_G(y) = n \log y - n \sum_{j \geq 1} a_j(G) y^{-2j} \quad (y > B), \quad (7)$$

$$[z^j] \log \left(\sum_k M_k z^k \right) = (-1)^{j+1} n a_j(G). \quad (8)$$

The second formula is a formal identity and computes the a_j by rational arithmetic without numerical roots. If $m = |E(G)|$ and $W = \sum_v \binom{d(v)}{2}$, then

$$a_1 = \frac{m}{n}, \quad a_2 = \frac{m/2 + W}{n}. \quad (9)$$

Indeed $M_2 = \binom{m}{2} - W$, so $na_2 = m^2/2 - M_2$.

Godsil's root-moment interpretation identifies $\sum_i \rho_i^{2j}$ with the sum over roots v of length- $2j$ returning walks in the rooted path tree of G at v [9]. Its vertices are simple paths starting at v , with adjacency by one-step extension or deletion. A returning walk of length $2j$ reaches depth at most j , so its contribution depends only on the radius- j neighborhood of v . These are path-tree walks, not arbitrary closed walks in G . This explains the finite-local-statistic nature of (5).

Lemma 2.2 (Global factorial moment domination). *For every integer $k \geq 0$,*

$$\mathbb{E}(X)_k = k! M_k \frac{I_{n-2k}}{I_n} \leq (\Delta/2)^k, \quad (10)$$

where the left side is zero when $2k > n$. Thus $|\mathbb{E}(1+w)^X| \leq \exp(\Delta|w|/2)$ for every $w \in \mathbb{C}$.

Proof. At the i th stage of an ordered matching-edge selection, at most $\Delta(n-2i)/2$ edges remain. Hence $k! M_k \leq \prod_{i=0}^{k-1} \Delta(n-2i)/2$. The sequence I_N is nondecreasing, and (1) gives $I_N \geq N I_{N-2}$ for $N \geq 2$. Iterating cancels the factors $n-2i$. The $k=0$ case is 1, including $\Delta=0$. Expanding the finite binomial-moment sum and taking absolute values proves the last assertion. $\square$

This bound is valid over the whole possible moment range. The main proof below uses direct analytic estimates; it does not infer a uniform infinite-tail estimate from fixed- k asymptotics.

3 The uniform expansion and its coefficient generator

Put $\varepsilon = n^{-1/2}$. In this section the numbers a_j always mean the actual statistics $a_j(G)$ at the current n .

Theorem 3.1 (Uniform complex PGF expansion). *There are universal rational polynomials $P_h(w; a_1, \dots, a_{\lfloor h/2 \rfloor + 1})$, with $P_0 = 1$ and degree at most h in w , such that, for each fixed integer $J \geq 0$ and real $R \geq 0$,*

$$\mathbb{E}(1+w)^X = e^{a_1 w} \left\{ \sum_{h=0}^J P_h(w; a(G)) \varepsilon^h + O_{\Delta, J, R}(\varepsilon^{J+1}) \right\} \quad (11)$$

uniformly over all G on n vertices of maximum degree at most Δ , and all $|w| \leq R$. No convergence or expansion of $a_j(G)$ is assumed.

The first coefficients are

$$\begin{aligned} P_1 &= -a_1 w, \\ P_2 &= \frac{3}{2} a_1 w + (\frac{3}{2} a_1^2 - a_2) w^2, \\ P_3 &= -\frac{11}{8} a_1 w + (2a_2 - 4a_1^2) w^2 + (a_1 a_2 - \frac{7}{6} a_1^3) w^3. \end{aligned} \quad (12)$$

Corollary 3.2 (Exponentially weighted signed errors). *Let*

$$q_{n,J}(\ell) = [z^\ell] \left[e^{a_1(z-1)} \sum_{h=0}^J P_h(z-1; a(G)) \varepsilon^h \right].$$

For every fixed $q \geq 0$,

$$\sum_{\ell \geq 0} q^\ell |\mathbb{P}(X = \ell) - q_{n,J}(\ell)| = O_{\Delta, J, q}(\varepsilon^{J+1}). \quad (13)$$

At $q = 0$, use $0^0 = 1$.

Proof. Choose a fixed $S > q$. On $|z| = S$, the PGF theorem with $w = z - 1$ bounds the analytic remainder by $C\varepsilon^{J+1}$. Cauchy's estimate bounds its ℓ th coefficient by $C\varepsilon^{J+1}S^{-\ell}$. The geometric sum is finite, giving (13). $\square$

The $q_{n,J}(\ell)$ may be negative. This is a signed asymptotic approximation, not a claim that truncation defines a probability law. For $J = 0$ it gives a uniform $O_\Delta(n^{-1/2})$ approximation to $\text{Poisson}(a_1(G))$ in total variation. If $a_1(G_n) \rightarrow \lambda$, this implies a Poisson limit. If those parameters oscillate, (11) remains valid but there need not be a single limiting law or constant-coefficient expansion. Nor does (13) assert a smooth all-orders expansion of total-variation distance through parameter values at which the relevant absolute values change branches, including integer Poisson means.

[Added 2 October 2026, batch 77.] For the path ($r = 1$) the limiting Poisson mean is the integer 1, precisely a case this caveat excludes from the general statement. Source 26 nevertheless proves an all-orders expansion of $d_{\text{TV}}(\mathcal{L}(X_n), \text{Poisson}(1))$ (Corollary 9.2), by a separate sign analysis of the constant-coefficient approximation: the deviation at $\ell = 1$ has $Q_1(1) = Q_2(1) = 0$ and $Q_3(1) = 2/3 > 0$. The two statements are consistent: the caveat concerns the general theorem, and the corollary settles one such case by an additional argument.

3.1 Two explicitly defined exterior sectors

Let $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$ and use the fixed bound B from (4). Define

$$\begin{aligned} K_G^\sigma &= \int_B^\infty \mu_G(y)\phi(y-\sigma) dy \quad (\sigma \in \{+1, -1\}), \\ C_G &= \int_{-B}^B \mu_G(y)\phi(y-1) dy. \end{aligned} \tag{14}$$

Both K_G^σ are nonnegative. Reflection and parity give the exact identity

$$A_G = K_G^+ + (-1)^n K_G^- + C_G, \quad |C_G| \leq B^n. \tag{15}$$

The bound follows from (6): on $[-B, B]$, each paired factor has magnitude at most B^2 and each zero-root factor at most B .

Theorem 3.3 (Separate all-orders sector estimates). *There are universal rational polynomials $D_h(a_1, \dots, a_{\lfloor h/2 \rfloor + 1})$, with $D_0 = 1$, such that for each fixed $J \geq 0$,*

$$K_G^\sigma = L_n^\sigma \left\{ \sum_{h=0}^J \sigma^h D_h(a(G)) \varepsilon^h + O_{\Delta, J}(\varepsilon^{J+1}) \right\}, \tag{16}$$

uniformly over the graph family and both signs, where

$$L_n^\sigma = \frac{1}{\sqrt{2}} \left(\frac{n}{e} \right)^{n/2} \exp \left(\sigma \sqrt{n} - \frac{1}{4} - a_1 \right). \tag{17}$$

The central term in (15) is smaller than either sector scale by every fixed inverse power of n .

In particular,

$$\begin{aligned} D_1 &= \frac{7}{24} + a_1, \\ D_2 &= -\frac{119}{1152} - \frac{29}{24}a_1 + \frac{3}{2}a_1^2 - a_2, \\ D_3 &= -\frac{7933}{414720} + \frac{961}{1152}a_1 - \frac{57}{16}a_1^2 + \frac{7}{6}a_1^3 + \frac{41}{24}a_2 - a_1a_2. \end{aligned} \tag{18}$$

The negative sector has its own relative remainder, even though it is smaller than the positive one by $e^{-2\sqrt{n}}$. The error from truncating the positive-sector series at a fixed algebraic order is much larger than the negative sector. Thus two-sector resolution refers to separate controlled integrals, not to numerical recovery of the smaller term from a single finitely truncated total asymptotic series.

Remark 3.4 (The cutoff is a convention). Changing B to another fixed common root bound changes the exact sectors by at most $O(C^n)$ for a fixed C . It does not change their algebraic asymptotic coefficients. We do not assert that the exact sector decomposition is canonically unique under arbitrary cutoff choices.

[Added 2 October 2026, batch 77.] *Two special cases elsewhere.* (i) For the empty graph ($r = 0$, $\mu_G(y) = y^n$, all $a_j = 0$), splitting at 0 instead of at B gives the two positive involution blocks A_σ of the repository's *Combinatorial Transseries Inverses* volume (`t2:eq:involution-A`), whose all-order expansion `t1:thm:ivforward` has the coefficients $D_h(0) = A_h$ and the factor σ^h of (16) [20]; the two cutoffs change the exact sectors by $O(C^n)$ only, as the remark above says. (ii) For the path ($r = 1$, $B = 2$), source 26 uses a different exact decomposition, through Chebyshev polynomials and the substitution $y = \lambda + \lambda^{-1}$; its sectors are $\mathcal{K}_n^\sigma = K_{P_n}^\sigma + \tau_n^\sigma$, with the same algebraic coefficients and a sharper, polynomial, central bound (Theorem 9.5).

3.2 An all-orders rational algorithm

Let $U \sim N(1/2, 1/2)$. For $h \geq 1$, set

$$p_h(u; a) = \frac{(-1)^{h+1} u^{h+2}}{h+2} - \sum_{j=1}^{\lfloor (h+2)/2 \rfloor} a_j (-1)^{h-2j+2} \binom{h+1}{h-2j+2} u^{h-2j+2}. \quad (19)$$

Define polynomial sequences by

$$b_0 = 1, \quad b_r = \frac{1}{r} \sum_{h=1}^r h p_h b_{r-h}, \quad D_r(a) = \mathbb{E} b_r(U; a). \quad (20)$$

All expectations are exact rational operations. The moments are generated by

$$\mu_0 = 1, \quad \mu_1 = \frac{1}{2}, \quad \mu_k = \frac{1}{2} \mu_{k-1} + \frac{k-1}{2} \mu_{k-2}.$$

For PGF coefficients put $A_r = D_r(0, \dots, 0)$ and

$$H_r(w; a) = D_r(-a_1 w, a_2 w^2, -a_3 w^3, \dots).$$

Then formal division gives

$$P_0 = 1, \quad P_r = H_r - \sum_{k=1}^r A_k P_{r-k}. \quad (21)$$

Equations (19)–(21) specify any prescribed finite order. They do not assert convergence of the infinite asymptotic series.

4 Uniform remainder proof

We prove the sector theorem first and then the complex PGF theorem. Constants may depend on the displayed fixed parameters, but never on the particular graph or its local statistics.

4.1 Local expansion at the positive saddle

For $y > B$, put $y = \sqrt{n} + u$ and use (7). Apart from the normalizing constant in (17), the integrand is the Gaussian $e^{-(u-\sigma/2)^2}$ times the exponential of

$$\sum_{h \geq 1} \frac{(-1)^{h+1} u^{h+2}}{h+2} \varepsilon^h - \sum_{j \geq 1} a_j \varepsilon^{2j-2} (1 + \varepsilon u)^{-2j} + a_1. \quad (22)$$

The remaining factor $\sqrt{\pi}/\sqrt{2\pi} = 1/\sqrt{2}$ gives exactly (17). On the window $|u| \leq n^{1/12}$, one has $|\varepsilon u| \leq n^{-5/12}$ and the nonconstant exponent is $O_\Delta(\varepsilon(1 + |u|^3)) = o(1)$.

For completeness, the following elementary estimate makes the uniformity explicit.

Lemma 4.1 (Gaussian window expansion). *For each fixed J , on $|u| \leq n^{1/12}$ the exponential of (22) equals*

$$\sum_{h=0}^J b_h(u; a) \varepsilon^h + \mathcal{R}_{n,J}(u; a), \quad |\mathcal{R}_{n,J}(u; a)| \leq C_{\Delta,J} \varepsilon^{J+1} (1 + |u|^N),$$

for some fixed integer $N = N(J)$ and all sufficiently large n uniformly in G . The same assertion holds when a_j is replaced by $(-1)^j a_j w^j$, uniformly for $|w| \leq R$, with constants also depending on R .

Proof. In the logarithmic series retain powers through ε^J ; Taylor's remainder for $\log(1 + \varepsilon u)$, after division by ε^2 , is bounded by $C_J \varepsilon^{J+1} |u|^{J+3}$ when $|\varepsilon u| \leq 1/2$. In the root sum it suffices to retain $j \leq \lfloor (J+2)/2 \rfloor$. Indeed (5) and $1 + \varepsilon u \geq 1/2$ bound the omitted j terms by a geometric series starting at order at least ε^{J+1} . Expanding each retained $(1 + \varepsilon u)^{-2j}$ gives a remainder bounded by $C_{\Delta, J} \varepsilon^{J+1} (1 + |u|^{J+1})$. This proves a polynomial-envelope remainder for the exponent, whose coefficients through order J are exactly (19).

Both the exact nonconstant exponent and its retained polynomial are uniformly $o(1)$ on this window. Taylor's formula for their exponentials therefore has a bounded exponential factor. Truncating the exponential of the retained polynomial by total ε degree gives (20). Terms of higher degree and the exponential Taylor remainder are bounded by $C \varepsilon^{J+1} (1 + |u|^N)$ for a fixed N , by finite multiplication and the estimate $O(\varepsilon(1 + |u|^3))$. The bounds for complex w follow by replacing the geometric-series ratio by a constant multiple depending on R . $\square$

Multiplying the remainder by $e^{-(u-\sigma/2)^2}$ and integrating gives $O(\varepsilon^{J+1})$, since every polynomial is integrable against that Gaussian. Extending the coefficient integrals from the local window to the whole real line costs a superalgebraically small amount. For $\sigma = +1$ this produces D_h . Each monomial in p_h has the parity of h in u , so $b_h(-u) = (-1)^h b_h(u)$. Reflection of $N(-1/2, 1/2)$ into $N(1/2, 1/2)$ gives $\sigma^h D_h$ for both signs.

4.2 Tails of the exterior integrals

For $y > B$, $0 \leq \mu_G(y) \leq y^n$. The exponent

$$f_\sigma(y) = n \log y - (y - \sigma)^2/2$$

has $f''_\sigma(y) = -n/y^2 - 1 \leq -1$ and maximizer $\sqrt{n} + O(1)$. On $y \geq \sqrt{n}/2$, strong concavity shows that removing $|y - \sqrt{n}| \leq n^{1/12}$ costs $O(\exp(-cn^{1/6}))$ relative to the saddle scale, uniformly for either sign. On $B \leq y \leq \sqrt{n}/2$, the crude bound $y^n \leq (\sqrt{n}/2)^n$ and interval length $O(\sqrt{n})$ give a relative bound of the form

$$O_\Delta \left(n^{1/2} (e/4)^{n/2} e^{O(\sqrt{n})} \right),$$

which is exponentially small. Finally B^n/L_n^σ has logarithm $-(n/2) \log n + O_\Delta(n)$, proving the central-error assertion. This completes Theorem 3.3.

4.3 Uniform complex marking

For real y and $|w| \leq R$, (6) gives

$$|F_{G,w}(y)| \leq (y^2 + RB^2)^{n/2}. \quad (23)$$

On $|y| \leq \sqrt{n}/2$ the integral is exponentially negligible relative to the positive saddle by the same $e/4$ comparison. On $|y| \geq \sqrt{n}/2$, the right side of (23) is at most $|y|^n \exp(2RB^2)$, so the previous Gaussian tail estimates apply. In the two local windows, $|w|\eta_i^2/y^2 < 1$ uniformly for large n , making the logarithmic expansion legitimate. The replacement is precisely

$$a_j \mapsto (-1)^j a_j w^j,$$

and the positive leading factor becomes $e^{a_1 w}$. Lemma 4.1 proves the uniform local expansion. The negative window is smaller than the positive scale by $e^{-2\sqrt{n}}$ times a uniformly bounded factor and hence is beyond every fixed algebraic order.

The corresponding unrestricted involution expansion is obtained by setting all $a_j = 0$. Its leading positive factor is nonzero and its series begins at 1. Dividing the marked expansion by it proves (11), with (21). The w degree bound follows because, after extracting $e^{a_1 w}$, an $a_1 w$ term first occurs at order ε , and a term $a_j w^j$ for $j \geq 2$ first occurs at order ε^{2j-2} , whose exponent is at least j . Products and division by a scalar series preserve the degree bound. This proves Theorem 3.1.

5 Fixed distance stabilization and A239144

Let S be a fixed nonempty finite set of positive integers and $r = \max S$. Define $G_n(S)$ on $[n]$ by $\{i, j\} \in E$ if $|i - j| \in S$. Its maximum degree is at most $2|S|$.

Theorem 5.1 (Exact eventual affine statistics). *For each fixed $j \geq 1$ there are effectively computable rational numbers $\alpha_j(S), \beta_j(S)$ such that*

$$a_j(G_n(S)) = \alpha_j(S) + \frac{\beta_j(S)}{n} \quad (n > jr). \quad (24)$$

Consequently every prescribed finite-order PGF or avoidance expansion for this family has constant coefficients after extracting its fixed leading exponential.

Proof. Introduce one variable x_e for each edge and the multivariate matching polynomial

$$Z_G(x) = \sum_{M \text{ matching}} \prod_{e \in M} x_e.$$

Its constant term is 1, so $\log Z_G$ is a formal power series. The coefficient of a monomial $\prod_e x_e^{m_e}$ depends only on the incompatibility graph of its support $H = \{e : m_e > 0\}$, namely the induced line graph on those edges: set every other variable to zero. If this incompatibility graph is disconnected, its matching polynomial factors over its components and its logarithm is a sum. A monomial involving multiple components therefore has zero coefficient.

Only supports connected in the line graph contribute to total degree j . Such a support has at most j edges, each of length at most r . Its largest and smallest vertices differ by at most jr : a connected chain of support edges joins them, and the sum of its edge lengths is at most jr . Translate its smallest vertex to zero. There are only finitely many resulting patterns P , including positive edge multiplicities totaling j . Let $\ell(P)$ be the span and $c(P)$ its rational logarithmic coefficient. When $n > jr$, the pattern occurs in exactly $n - \ell(P)$ translates. Therefore

$$[z^j] \log \left(\sum_k M_k(G_n(S)) z^k \right) = \sum_P c(P)(n - \ell(P)) = u_j n + v_j. \quad (25)$$

Setting every $x_e = z$ proves the first equality; (8) proves (24). The finite pattern sum is an algorithm. Alternatively, after the theorem has justified affine dependence, two consecutive exact evaluations at $n > jr$ recover u_j, v_j ; a third is a useful implementation check. Substituting (24) with $1/n = \varepsilon^2$ into the universal generator proves the constant-coefficient conclusion. $\square$

For $S = \emptyset$ the graph is empty and the assertion is immediate. The proof is structural; observing empirical linearity at a few values would not be a substitute.

5.1 First path power corrections

For $S = \{1, \dots, r\}$, write $G_n = P_n^r$. For sufficiently large n ,

$$m = rn - \frac{r(r+1)}{2}, \quad \frac{W}{n} = r(2r-1) + O_r(n^{-1}),$$

so

$$a_1 = r - \frac{r(r+1)}{2n}, \quad a_2 = 2r^2 - \frac{r}{2} + O_r(n^{-1}).$$

Insert these in (12) at $w = -1$, including the factor e^{-a_1} , to obtain (2). The $n^{-3/2}$ coefficients inside the square brackets are

$$r = 1 : \frac{1}{24}, \quad r = 2 : -\frac{11}{12}, \quad r = 3 : -\frac{39}{8}.$$

For the full marked distribution, the first two constant-parameter corrections are

$$\mathbb{E}(1+w)^X = e^{rw} \left[1 - rw\varepsilon + \left\{ \frac{r(2-r)}{2}w - \frac{r(r-1)}{2}w^2 \right\} \varepsilon^2 + O_{r,R}(\varepsilon^3) \right] \quad (26)$$

uniformly for $|w| \leq R$.

Taking formal logarithms of the positive sector (the other terms being superalgebraically smaller), define

$$L_{r,J}(x) = \frac{x}{2}(\log x - 1) + \sqrt{x} - \frac{\log 2}{2} - \frac{1}{4} - r + \sum_{j=1}^J \gamma_{r,j} x^{-j/2}, \quad (27)$$

$$\log A_{n,r} = L_{r,J}(n) + O_{r,J}(n^{-(J+1)/2}). \quad (28)$$

The first coefficients are

$$\gamma_{r,1} = r + \frac{7}{24}, \quad \gamma_{r,2} = -\frac{r(r+1)}{2} - \frac{7}{48}, \quad \gamma_{r,3} = r^2 - \frac{r}{8} + \frac{37}{1920}. \quad (29)$$

For $r = 1$ these are $31/24, -55/48, 1717/1920$. In particular the first absolute correction agrees with the previously recorded Kotěšovec coefficient. For $r = 0$ they recover the unrestricted-involution coefficients. The companion scripts give exact coefficients through order six for $r = 0, 1, 2, 3$; the recurrence, rather than a table truncated at six, is the all-orders specification.

[Added 2 October 2026, batch 77.] At $r = 1$ source 26 computes the same objects independently: its logarithmic coefficients are $\beta_j = \gamma_{1,j}$ (it adds $\beta_4 = -75/128$, and $\beta_5, \dots, \beta_8$ in Appendix A), its constant $c_A = -\frac{\log 2}{2} - \frac{5}{4}$ is the constant term of $L_{1,J}$, and its polynomials $P_1(w) = -w$, $P_2(w) = w/2$ are (26) at $r = 1$. For $r = 0$ the values $\gamma_{0,1}, \gamma_{0,2}, \gamma_{0,3} = 7/24, -7/48, 37/1920$ are source 26's $\lambda_1, \lambda_2, \lambda_3$ (Lemma 9.3).

6 Three inverse conventions

An integer sequence has no unique smooth inverse. We keep three conventions separate and restrict all claims here to fixed r . The moving-statistic theorem for arbitrary graph sequences by itself does not imply any comparable inverse theorem.

6.1 The smooth asymptotic root

Fix r, J . Since

$$L'_{r,J}(x) = \frac{1}{2} \log x + O_r(x^{-1/2}) > 0$$

for all sufficiently large x , the equation $L_{r,J}(x) = \log y$ has a unique sufficiently large root, denoted $x_{r,J}(y)$. It is an asymptotic convention, not an asserted exact analytic continuation of $A_{n,r}$. From (28) and the mean-value theorem,

$$x_{r,J}(A_{n,r}) = n + O_{r,J}\left(\frac{n^{-(J+1)/2}}{\log n}\right). \quad (30)$$

There is a finite explicit algorithm reaching this precision. Put $b = \log y$, use the positive real branch of Lambert W , and set

$$x_0 = \frac{2b}{W(2b/e)}, \quad x_{k+1} = x_k - \frac{L_{r,J}(x_k) - b}{L'_{r,J}(x_k)}. \quad (31)$$

The seed solves $x(\log x - 1)/2 = b$ exactly, while $x_{r,J}(y) - x_0 = O_r(\sqrt{x_0}/\log x_0)$. On that neighborhood $L' = \Theta(\log x_0)$ and $L'' = O(1/x_0)$. If $e_k = x_k - x_{r,J}(y)$, Taylor's theorem gives

$$e_{k+1} = O_{r,J}\left(\frac{e_k^2}{x_0 \log x_0}\right), \quad e_k = O_{r,J,k}\left(\frac{x_0^{1-2^{k-1}}}{(\log x_0)^{2^{k+1}-1}}\right).$$

For a fixed desired order, $k = \lceil \log_2(J+3) \rceil$ updates therefore give error $o(x_0^{-(J+1)/2}/\log x_0)$, smaller than the intrinsic asymptotic-log error. The number of updates need not grow with y .

6.2 The integer threshold

Define

$$N_r(y) = \min\{n \geq 0 : A_{n,r} \geq y\}.$$

Adjoining a fixed terminal label embeds the objects on $[n-1]$ into those on $[n]$. For $n \geq r+2$, the transposition $(1, n)$ with every other vertex fixed is an additional allowed object, so the sequence is strictly increasing from that point.

Proposition 6.1 (Two ceiling enclosure). *For fixed r, J there are constants $C_{r,J}, y_{r,J} > 0$ such that, for $y \geq y_{r,J}$, if $x = x_{r,J}(y)$ and $\delta = C_{r,J}x^{-(J+1)/2}/\log x$, then*

$$\lceil x - \delta \rceil \leq N_r(y) \leq \lceil x + \delta \rceil. \quad (32)$$

A finite Newton approximation from (31) can replace x after enlarging the constant.

Proof. Write $\alpha = (J+1)/2$ and bound the error in (28) by $B_0 m^{-\alpha}$. In a fixed neighborhood of x , take $L' \geq (\log x)/4$ and $m^{-\alpha} \leq 2^\alpha x^{-\alpha}$. Choose $C > 4 \cdot 2^\alpha B_0$. At $m_- = \lceil x - \delta \rceil - 1 < x - \delta$ and $m_+ = \lceil x + \delta \rceil \geq x + \delta$, the mean-value theorem implies $\log A_{m_-,r} < \log y < \log A_{m_+,r}$ for sufficiently large y . Eventual strict monotonicity gives (32). $\square$

The constants are asserted to exist; they are not supplied as certified numerical bounds at explicit finite inputs. There is no unconditional single-ceiling formula, because y may be arbitrarily close to an exact sequence value.

6.3 The exact log linear convention

For $x = m + \theta$ in the eventually increasing range, $0 \leq \theta \leq 1$, define

$$\begin{aligned}\log \mathcal{A}_r(m + \theta) &= (1 - \theta) \log A_{m,r} + \theta \log A_{m+1,r}, \\ \tilde{L}_{r,J}(m + \theta) &= (1 - \theta) L_{r,J}(m) + \theta L_{r,J}(m + 1).\end{aligned}$$

This specifies an actual interpolation convention for the original sequence. If the consecutive values $L_{r,J}(m), L_{r,J}(m + 1)$ bracket $\log y$, its approximate chord inverse is

$$\tilde{x}_{r,J}(y) = m + \frac{\log y - L_{r,J}(m)}{L_{r,J}(m + 1) - L_{r,J}(m)}. \quad (33)$$

Endpoint errors in (28) extend uniformly by convex interpolation; both interpolants have slopes $\Theta(\log x)$. It follows, including across a knot by their local lower slope bound, that

$$\mathcal{A}_r^{-1}(y) = \tilde{x}_{r,J}(y) + O_{r,J} \left(\frac{\tilde{x}_{r,J}(y)^{-(J+1)/2}}{\log \tilde{x}_{r,J}(y)} \right). \quad (34)$$

The interpolation defect of the smooth L is $O(\sup |L''|) = O(1/x)$. Thus the smooth and chord roots generally differ by $O(1/(x \log x))$. One cannot silently identify these two inverses at arbitrary order; the chord construction is required for the exact log-linear convention.

[Added 2 October 2026, batch 77.] *Relation to the repository's inversion theorems; no novelty is claimed for the smooth inverse.* (i) The seed of (31) solves $\frac{x}{2}(\log x - 1) = b$; in the variable $V = \log x - 1$ this is $V + \log V = \log(2b/e)$, an instance of the Lambert core, Theorem `p0:thm:lambert-core` of *Transseries and Inversion* [19] (both of its coefficients equal to 1, branch W_0). For $r = 0$ the all-orders inverse of the dominant involution block, including the unbounded first correction of order $\sqrt{x_0}/\log x_0$ seen above, is Theorem `t1:thm:ivinverse` of *Combinatorial Transseries Inverses* [20]; for $r \geq 1$ only the constant term $-r$ and the coefficients $\gamma_{r,j}$ of $L_{r,J}$ change. The reversion machinery around a Lambert core is Theorem `p0:thm:lambert-centered` of [19] (for gamma-quotient phases, `t2:thm:balanced-inverse` of [20]); the finite Newton iteration above is a different, equally finite, way to reach a prescribed order. (ii) The log-linear interpolant $\mathcal{A}_r$ is an admissible interpolation in the sense of [19], so part (1) of Theorem `p0:thm:staircase` gives exactly $N_r(y) = \lceil \mathcal{A}_r^{-1}(y) \rceil$; combined with (34) this yields a two-ceiling enclosure around the chord root $\tilde{x}_{r,J}(y)$, and part (2), the separation condition, is the reason no unconditional single-ceiling formula is available. Specific to this family are only the eventual monotonicity from $n \geq r + 2$, the stabilized coefficients $\gamma_{r,j}$ feeding $L_{r,J}$, and the enclosures as stated. At $r = 1$ source 26 states the smooth root, the Newton algorithm and the two-ceiling enclosure in the same form (Section 9.9).

7 Reproducible checks and limitations

The companion archive contains the article source, a build script, a portable mathematical replay, exact coefficient generation, independent symbolic and finite-graph tests, and reference outputs. It contains no third-party full papers. Bibliographic links identify the sources without redistributing them.

[Added 2 October 2026, batch 77.] In this report source 19's scripts are shipped as `code/19-distance-*` and its six reference outputs as `data/19-distance-expected-*.json`; its delivered PDF and its checksum manifest `MANIFEST.json` are not shipped. Source 26's checks are described in Section 9.10. The README

of this report gives rerun instructions for the shipped names; both suites were replayed on copies at intake with every check passing.

The producer calculation uses the recurrence (20), Gaussian moment recursion, and a finite-frontier matching counter. The independent calculation instead expands the exponent directly, forms a product of truncated exponential series, uses a closed Gaussian moment formula, and divides by a finite geometric inverse. All universal sector and PGF coefficients through order six agree symbolically. A vertex-subset recurrence independently checks 120 path-power matching-count cases, including direct complement counts, for $0 \leq n \leq 14$. Exact factorial-moment inequalities and (9) pass for 1,199 graphs: all simple graphs with $1 \leq n \leq 5$, plus 100 seeded random graphs of orders 6 through 10.

Fixed-range tests verify 100 exact local-statistic values across $r = 1, \dots, 5$, $j = 1, \dots, 4$, starting immediately above $n = jr$. The separate inverse tests use the exact affine statistics, compute logarithmic coefficients through order six for $r = 0, 1, 2, 3$, and check smooth roots at sequence nodes and chord roots at fractional indices. Numerical sector tests for disjoint triangles check both signs at $n = 60, 150, 300, 600$.

The following values illustrate the order-six avoidance approximation. Here E_6 is exact $A_{n,r}/I_n$ minus the truncation of (11) through $h = 6$ with the actual graph statistics. Displayed decimals are rounded; the archive preserves higher-precision data.

r	n	$A_{n,r}/I_n$	$n^{7/2}E_6$
1	100	0.402851003599	-1.83708
1	500	0.383965373246	-1.92108
2	100	0.160924606641	-3.38493
2	500	0.147158392484	-3.24595
3	100	0.0637215724335	1.11376
3	500	0.0562953604037	2.07702

These checks corroborate the derivation; they are not proofs of uniform asymptotic remainders or effective error constants. The mathematical proof is analytic and finite-order. It does not certify a convergent expansion, growing-degree uniformity, or numerical threshold enclosures at an explicit starting value. The independent review is human-readable mathematical review supported by exact computation, not formal proof-assistant verification.

8 Questions opened by the refinement

Several next questions are substantive while remaining distinct from the established results.

1. **Growing forbidden degree.** What explicit relation between $\Delta(n)$, the requested order, and n preserves a useful complex-PGF expansion? The geometric root-tail control and the local exponential estimates expose where fixed-degree constants enter. McLeod's wider degree range in the regular perfect-matching model is a relevant benchmark, not an automatic extension.
2. **Effective error constants.** Can the Gaussian-window and tail estimates be optimized into explicit constants and a computable validity threshold, sufficient to certify particular integer inversions?

3. **Efficient fixed-distance coefficients.** How can connected-support enumeration, transfer matrices, or path-tree recursions compute $\alpha_j(S), \beta_j(S)$ efficiently for large order or nonconsecutive distance sets? The proof gives an effective finite procedure but no optimal complexity bound.
4. **Several overlapping forbidden classes.** Can the same root-based analytic control give multivariate all-orders weighted laws for overlapping edge classes? Existing joint perfect-matching Poisson results provide a natural comparison, but the single-variable matching-root factorization used here does not immediately settle that problem.
5. **Exponentially improved approximation.** What controlled truncation or resummation recovers the smaller sector from the total count at a precision beyond every fixed algebraic order? The exact two-integral definition alone does not answer that numerical question.

Before making a publication-level novelty claim, a direct comparison with the final McLeod journal version and the full Wanless paper remains appropriate. The article’s mathematical statements and its usefulness as a reproducible refinement do not depend on a claim of priority.

[Added 2 October 2026, batch 77.] Source 26’s six questions for the adjacent case are printed in Section 9.11. Its effective-constants question is question 2 above specialized to $r = 1$; its small-sector question is question 5 with an added large-order question for d_j ; its joint-law question is a different refinement from question 4. Its question on other sparse forbidden graphs is largely answered by this report’s Theorems 3.1, 3.3 and 5.1 and is re-scoped there.

9 Adjacent transpositions ($r = 1$): the full law of A217876

[Added 2 October 2026, batch 77.] *Merge note.* This section prints source 26, *All order asymptotics for adjacent transpositions in involutions* (a research note on OEIS A170941 and A217876). Its Theorem 9.1, the avoidance expansions of Section 9.7 and its sector expansion (60) are the case $G = P_n$ (forbidden distance 1, so $\Delta = 2$ and $B = 2$) of Theorems 3.1 and 3.3 and Corollary 3.2, rewritten with constant coefficients by Theorem 5.1; the coefficients agree (Section 5.1). Printed here are what source 26 says beyond those theorems: the adjacent-transposition law with the polynomials $Q_j(\ell)$, the total-variation corollary, the exact moments and triangle, a second proof of the law (Sections 9.4–9.6, by growing-moment summation instead of matching roots), the exact Chebyshev sectors with their residual bound and sector ratio, the log-linear inverse, the source’s checks and questions, and its coefficients through order eight (Appendix A). Its smooth root, Newton algorithm and two-ceiling enclosure are Section 6 at $r = 1$ and are given by pointer in Section 9.9. Notation follows the table of the provenance section; in this section $r = 1$ throughout, and the letters renamed from source 26 are listed there.

9.1 The model and the scope of the result

An involution consists of fixed points and disjoint transpositions. Equivalently, it is a matching on the complete graph on $[n] = \{1, \dots, n\}$, with unmatched vertices interpreted as fixed points. With I_n as in (1) (OEIS A000085 [4]), for a uniformly chosen involution define

$$X_n = \#\{i \in \{1, \dots, n-1\} : (i, i+1) \text{ is a transposition}\},$$

so $X_n = X_{P_n}$ in the notation of Section 2. Write $p_n(\ell) = \mathbb{P}(X_n = \ell)$ and $\mathcal{G}_n(z) = \mathbb{E}z^{X_n}$. Then

$$T^{\text{adj}}(n, \ell) = I_n p_n(\ell) \quad \text{is OEIS A217876 [3],} \quad a_n = T^{\text{adj}}(n, 0) = A_{n,1} \quad \text{is OEIS A170941.}$$

Thus a_n also counts matchings in the complement of the path P_n . For example, $a_3 = 2$: the identity and the transposition $(1, 3)$ are allowed. The first values are 1, 1, 1, 2, 5, 13, 37, 112, 363. In the indexing used here, the avoidance recurrence of Haslinger and Stadler (Section 1.1) is

$$a_n = a_{n-1} + (n-1)a_{n-2} - a_{n-3} + a_{n-4} \quad (n \geq 4), \quad (a_0, a_1, a_2, a_3) = (1, 1, 1, 2). \quad (35)$$

This is a regular combinatorial model with a direct finite-size question: how accurately does a Poisson law describe the number of forbidden path edges? The matching interpretation makes exact sampling and recurrence checks straightforward. The historical RNA interpretation concerns an idealized contact-diagram count with nearest-neighbor contacts excluded. Crossing arcs are allowed in this model. It is not, by itself, a model of chemical base compatibility, energy, or the usual noncrossing secondary-structure ensemble.

Conventions. Throughout, asymptotics are for integers $n \rightarrow \infty$ unless a real argument is explicitly introduced. Put $\varepsilon = n^{-1/2}$ and $w = z - 1$. A fixed-order expansion means: for each fixed truncation order there is a remainder bound, with constants allowed to depend on that order and on the explicitly fixed compact set or weight. It does not assert convergence of the infinite formal series, nor uniformity when the order grows with n .

9.2 The full probability law

Theorem 9.1 (Uniform generating function and weighted law). *There are effectively computable rational polynomials $P_j(w)$, with*

$$P_0 = 1, \quad P_j(0) = 0 \quad (j \geq 1), \quad \deg P_j \leq j,$$

such that, for every fixed integer $J \geq 0$ and every fixed $R < \infty$,

$$\mathcal{G}_n(1+w) = e^w \sum_{j=0}^J P_j(w) \varepsilon^j + O_{J,R}(\varepsilon^{J+1}), \quad |w| \leq R. \quad (36)$$

Let

$$Q_j(\ell) = \ell! [z^\ell] e^z P_j(z-1), \quad \ell \geq 0. \quad (37)$$

Each Q_j is a rational polynomial in ℓ of degree at most j . For every fixed $q \geq 0$,

$$\sum_{\ell \geq 0} q^\ell \left| p_n(\ell) - \frac{e^{-1}}{\ell!} \sum_{j=0}^J Q_j(\ell) \varepsilon^j \right| = O_{J,q}(\varepsilon^{J+1}). \quad (38)$$

For $q = 0$, the sum means just its $\ell = 0$ term.

The first generating-function polynomials are

$$\begin{aligned} P_1(w) &= -w, & P_2(w) &= \frac{w}{2}, \\ P_3(w) &= \frac{w^3}{3} - \frac{3w}{8}, & P_4(w) &= -\frac{w^4}{12} - \frac{w^3}{2} + \frac{w}{8}, \\ P_5(w) &= -\frac{w^5}{20} + \frac{w^4}{6} + \frac{9w^3}{8} + \frac{w}{128}. \end{aligned}$$

Correspondingly,

$$\begin{aligned} Q_0(\ell) &= 1, & Q_1(\ell) &= 1 - \ell, & Q_2(\ell) &= \frac{\ell - 1}{2}, \\ Q_3(\ell) &= \frac{\ell^3}{3} - 2\ell^2 + \frac{55\ell}{24} + \frac{1}{24}. \end{aligned}$$

For $J \geq 1$, the truncated Poisson-polynomial approximation is a signed measure, not necessarily a nonnegative probability law at all ℓ . It has total mass one. The norm statement remains valid on the entire nonnegative integer axis, including beyond the exact support $\ell \leq \lfloor n/2 \rfloor$.

[Added 2 October 2026, batch 77.] *First proof.* Theorem 9.1 is Theorem 3.1 and Corollary 3.2 for $G = P_n$, after Theorem 5.1 ($a_1 = 1 - 1/n$, $a_2 = 3/2 + O(n^{-1})$ and so on) has been substituted and the factor $e^{a_1 w} = e^w e^{-w\varepsilon^2}$ re-expanded; this is how (26) arises. Hence $P_j(w)$ here is *not* $P_j(w; a(P_n))$ of Theorem 3.1: for instance $P_2(w; a(P_n)) = \frac{3}{2}a_1 w + (\frac{3}{2}a_1^2 - a_2)w^2 \rightarrow \frac{3}{2}w$, while $P_2(w) = w/2$. Source 26's own proof, by a different route, occupies Sections 9.4–9.6.

Corollary 9.2 (Avoidance and total variation). *For each fixed J ,*

$$\frac{a_n}{I_n} = e^{-1} \sum_{j=0}^J P_j(-1) \varepsilon^j + O_J(\varepsilon^{J+1}), \quad (39)$$

$$d_{\text{TV}}(\mathcal{L}(X_n), \text{Poisson}(1)) = e^{-1} \sum_{j=1}^J V_j \varepsilon^j + O_J(\varepsilon^{J+1}), \quad (40)$$

$$V_j = 2P_j(-1) + P'_j(-1).$$

In particular,

$$\begin{aligned} \frac{a_n}{I_n} &= e^{-1} \left(1 + \varepsilon - \frac{\varepsilon^2}{2} + \frac{\varepsilon^3}{24} + \frac{7\varepsilon^4}{24} - \frac{1759\varepsilon^5}{1920} + O(\varepsilon^6) \right), \\ d_{\text{TV}}(\mathcal{L}(X_n), \text{Poisson}(1)) &= e^{-1} \left(\varepsilon - \frac{\varepsilon^2}{2} + \frac{17\varepsilon^3}{24} - \frac{11\varepsilon^4}{24} + \frac{1217\varepsilon^5}{1920} + O(\varepsilon^6) \right). \end{aligned}$$

Here total variation is one half of the ℓ^1 distance between the probability masses.

The avoidance ratio agrees with (2) at $r = 1$ ($1 + \varepsilon - \varepsilon^2/2 + \varepsilon^3/24$).

9.3 Exact moments and their global bound

A set of k specified adjacent transpositions must be a k -matching in P_n . The number of such sets is $\binom{n-k}{k}$: if the left endpoints are $i_1 < \dots < i_k$ with $i_{s+1} \geq i_s + 2$, map them to the ordinary k -subset $\{i_s - (s-1)\}_{s=1}^k$ of $[n-k]$. The remaining labels carry any involution. Thus the exact binomial moments are

$$m_{n,k} := \mathbb{E} \binom{X_n}{k} = \binom{n-k}{k} \frac{I_{n-2k}}{I_n}, \quad \mathcal{G}_n(1+w) = \sum_{k \geq 0} m_{n,k} w^k, \quad (41)$$

where $m_{n,k} = 0$ if $2k > n$. This is (3) with $M_k(P_n) = \binom{n-k}{k}$. The unnormalized selected-pair count already appears in Ganjtabesh and Steyaert's equation (17), with its avoidance inclusion–exclusion

specialization in equation (18) [14]. Here it is explicitly interpreted as a binomial moment; Li uses the factorial-moment mechanism for the full Poisson law.

The recurrence for I_n supplies a useful bound valid for every k , not only fixed k :

$$0 \leq m_{n,k} \leq \frac{1}{k!}. \quad (42)$$

This is Lemma 2.2 with $\Delta = 2$. Source 26 proves it directly:

$$I_n \geq \prod_{s=0}^{k-1} (n-1-2s)I_{n-2k}, \quad k! \binom{n-k}{k} = \prod_{s=0}^{k-1} (n-k-s),$$

and $n-k-s \leq n-1-2s$ for $0 \leq s \leq k-1$. For $k=0$ the products are empty. This factorial domination controls the exact tail of the PGF uniformly in n .

Coefficient extraction in (41) also gives a finite exact formula for the whole triangle:

$$T^{\text{adj}}(n, \ell) = \sum_{k=\ell}^{\lfloor n/2 \rfloor} (-1)^{k-\ell} \binom{k}{\ell} \binom{n-k}{k} I_{n-2k}. \quad (43)$$

It is useful for exact checks but can involve severe cancellation in floating-point arithmetic.

9.4 Second proof, step one: a scalar expansion from a Gaussian integral

Source 26 gives the remainder proof explicitly so that the later uniformity argument rests on more than formal coefficient manipulation. The exponential generating function of involutions is $\exp(z + z^2/2)$. Comparing it with the moment generating function of $1 + Z$, where Z is standard normal, yields

$$I_n = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} y^n e^{-(y-1)^2/2} dy. \quad (44)$$

Let $\mathcal{J}_n^+$ be the integral over $y > 0$, including the normalization, and $\mathcal{J}_n^-$ the absolute reflected integral over $y < 0$. Then $I_n = \mathcal{J}_n^+ + (-1)^n \mathcal{J}_n^-$. The negative reflected integrand is e^{-2y} times the positive integrand. (These are the blocks $A_\sigma(n)$ of [20, t2:eq:involution-A].)

The positive log-integrand

$$h_n(y) = n \log y - \frac{(y-1)^2}{2}$$

has its unique maximum at $y_* = (1 + \sqrt{1+4n})/2$, with $h_n''(y) \leq -1$. On a fixed interval about y_* the second derivative is also bounded below by a negative constant, so the total positive integral is bounded below by a constant times $e^{h_n(y_*)}$. Strong concavity bounds its portion with $y \leq \sqrt{n}/2$ by $O(e^{-cn})$ of the total. Splitting the negative integral at that point gives

$$\frac{\mathcal{J}_n^-}{\mathcal{J}_n^+} \leq e^{-\sqrt{n}} + O(e^{-cn}). \quad (45)$$

Parity therefore has no effect on any fixed algebraic order.

Set $y = \sqrt{n} + u$. The exponent becomes

$$\frac{n}{2}(\log n - 1) + \sqrt{n} - \frac{1}{2} - u^2 + u + \mathcal{F}(\varepsilon, u), \quad \mathcal{F}(\varepsilon, u) = \sum_{h \geq 1} \frac{(-1)^{h+3}}{h+2} u^{h+2} \varepsilon^h.$$

The series here is used locally, with its removable analytic interpretation at $\varepsilon = 0$. Let U have density $\pi^{-1/2}e^{-(u-1/2)^2}$, that is, mean $1/2$ and variance $1/2$ (the U of (20)), and define

$$A_j = \mathbb{E}([\varepsilon^j]e^{\mathcal{F}(\varepsilon, U)}), \quad \lambda_j = [\varepsilon^j] \log \left(\sum_{m \geq 0} A_m \varepsilon^m \right). \quad (46)$$

Each coefficient is a finite polynomial Gaussian expectation. It does not require convergence of the infinite asymptotic series. ($A_j = D_j(0, \dots, 0)$, as in Section 3.)

Lemma 9.3 (Scalar remainder). *For every fixed $J \geq 0$,*

$$I_n = \frac{e^{\frac{n}{2}(\log n - 1) + \sqrt{n} - 1/4}}{\sqrt{2}} \left(\sum_{j=0}^J A_j \varepsilon^j + O_J(\varepsilon^{J+1}) \right), \quad (47)$$

$$\log I_n = \frac{n}{2}(\log n - 1) + \sqrt{n} - \frac{\log 2}{2} - \frac{1}{4} + \sum_{j=1}^J \lambda_j \varepsilon^j + O_J(\varepsilon^{J+1}). \quad (48)$$

The first logarithmic coefficients are $\lambda_1 = 7/24$, $\lambda_2 = -7/48$, $\lambda_3 = 37/1920$, and $\lambda_4 = 5/128$.

Proof. Restrict first to $|u| \leq \varepsilon^{-1/6} = n^{1/12}$. For $0 \leq s \leq \varepsilon$ one has $|su| \leq \varepsilon^{5/6}$, and for each fixed $m \geq 1$,

$$|\mathcal{F}(s, u)| \leq Cs|u|^3, \quad |\partial_s^m \mathcal{F}(s, u)| \leq C_m |u|^{m+2}.$$

Consequently $e^{\mathcal{F}(s, u)}$ stays bounded and its m th derivative is bounded by $C_m(1 + |u|)^{3m}$. Taylor's theorem bounds the remainder after degree J by

$$C_J \varepsilon^{J+1} (1 + |u|)^{3(J+1)}.$$

Multiplication by e^{-u^2+u} gives an integrable majorant independent of n . Strong concavity about $y_* = \sqrt{n} + 1/2 + O(\varepsilon)$ makes the discarded positive-integral tail $O(e^{-cn^{1/6}})$ relative to the main normalization. Extending each polynomial–Gaussian integral to $\mathbb{R}$ has the same negligible effect. Completing the square supplies $e^{-1/4}/\sqrt{2}$, and (45) absorbs the negative part. This proves (47); its leading coefficient is one, so taking logarithms proves (48).

Rationality follows from the Gaussian moment recursion

$$\mu_0 = 1, \quad \mu_1 = \frac{1}{2}, \quad \mu_m = \frac{\mu_{m-1}}{2} + \frac{m-1}{2} \mu_{m-2}.$$

□

Lemma 9.3 is the case $r = 0$ of (28) ($\lambda_j = \gamma_{0,j}$) and, for the exact blocks, the case $\sigma = +1$ of [20, t1:thm:ivforward].

9.5 Second proof, step two: the growing moment estimate

The scalar remainder can be transferred to a range of k growing with n . Define the formal series

$$\begin{aligned} \mathcal{H}(k, \varepsilon) = & - \sum_{m \geq 1} \frac{\varepsilon^{2m}}{m} \sum_{i=0}^{k-1} (k+i)^m \\ & + \frac{(1 - 2k\varepsilon^2) \log(1 - 2k\varepsilon^2) + 2k\varepsilon^2}{2\varepsilon^2} + \frac{\sqrt{1 - 2k\varepsilon^2} - 1}{\varepsilon} \\ & + \sum_{h \geq 1} \lambda_h \varepsilon^h \left((1 - 2k\varepsilon^2)^{-h/2} - 1 \right), \end{aligned} \quad (49)$$

where the apparent singularities at $\varepsilon = 0$ are removable, and set

$$C_j(k) = [\varepsilon^j] e^{\mathcal{H}(k, \varepsilon)}. \quad (50)$$

The coefficient of ε^j in $\mathcal{H}$ is a polynomial in k of degree at most j . For the four types of terms, respectively, the degrees at orders $2m$, $2m$, $2h - 1$, and $\ell + 2h$ are at most $m + 1$, $m + 1$, h , and h . Exponentiation preserves the degree bound. In particular

$$C_0 = 1, \quad C_1 = -k, \quad C_2 = \frac{k}{2}, \quad C_3 = \frac{k^3}{3} - k^2 + \frac{7k}{24}.$$

Lemma 9.4 (Factorially summable uniform error). *For each fixed $J \geq 0$ there are constants c_J, N_J such that, whenever $n \geq N_J$ and $k \geq 0$ is an integer satisfying $(1 + k)/\sqrt{n} \leq 1/8$,*

$$\left| m_{n,k} - \frac{1}{k!} \sum_{j=0}^J C_j(k) \varepsilon^j \right| \leq \frac{c_J}{k!} (1 + k)^{J+1} \varepsilon^{J+1}. \quad (51)$$

Proof. In (49), retain only λ_h with $h \leq J$ and replace the first formal series by its exact analytic form $\sum_{i=0}^{k-1} \log(1 - (k+i)\varepsilon^2)$. Call the resulting germ $\mathcal{F}_J(k, \varepsilon)$. Direct subtraction of (48) at $n - 2k$ and n gives

$$\log(k! m_{n,k}) = \mathcal{F}_J(k, \varepsilon) + E_{n,k}, \quad |E_{n,k}| \leq c'_J \varepsilon^{J+1}.$$

The scalar errors can be subtracted separately because $n - 2k \geq n/2$ in this range. No derivative of an unknown remainder is used. The $-k \log n$ term from the scalar difference cancels $k \log n$ from the binomial coefficient.

For $\vartheta_k = 1/(4(1 + k))$, all logarithms and powers in $\mathcal{F}_J$ are analytic on $|\varepsilon| \leq \vartheta_k$, on branches agreeing with their value at zero. On this disk $|2k\varepsilon^2| \leq 1/32$. Elementary power-series bounds give, uniformly in k ,

$$|\mathcal{F}_J(k, \varepsilon)| \leq \Omega_J := \frac{1}{2} + \frac{5}{31} + \sum_{h=1}^J |\lambda_h| 4^{-h} \left(\left(\frac{32}{31} \right)^{h/2} + 1 \right).$$

For example the bounds on the first three terms are $3/31$, $2/31$, and $1/2$. Cauchy's estimate and a geometric tail therefore give, for $|\varepsilon| \leq \vartheta_k/2$,

$$\left| e^{\mathcal{F}_J(k, \varepsilon)} - \sum_{j=0}^J C_j(k) \varepsilon^j \right| \leq 2e^{\Omega_J} (4(1 + k)|\varepsilon|)^{J+1}.$$

Terms involving λ_h first appear at degree $h + 2$, so the displayed coefficients do not depend on the truncation choice. Multiplying by $e^{E_{n,k}}$ adds at most $e^{\Omega_J+1} c'_J \varepsilon^{J+1}$ once n is sufficiently large. Divide by $k!$ and absorb constants. $\square$

9.6 Second proof, step three: summation and probability coefficients

Second proof of Theorem 9.1 (source 26). Fix J, R and take

$$k_0 = \lceil (J + 2) \log n / \log \log n \rceil.$$

The growing-moment lemma applies for $k \leq k_0$ once n is large. Summing its error on $|w| \leq R$ gives at most

$$c_J \varepsilon^{J+1} \sum_{k \geq 0} \frac{(1 + k)^{J+1} R^k}{k!} = O_{J,R}(\varepsilon^{J+1}).$$

The exact tail is bounded by (42). The coefficient-approximation tail has only an additional fixed polynomial factor in k . Stirling's bound gives, for each fixed d ,

$$\sum_{k > k_0} \frac{(1+k)^d R^k}{k!} = n^{-(J+2)+o(1)} = o(\varepsilon^{J+1}).$$

Thus the sum can be extended to all k . If $\mathcal{T}_m$ denotes the Touchard polynomial determined by $\sum_{k \geq 0} k^m w^k / k! = e^w \mathcal{T}_m(w)$, define

$$P_j(w) = e^{-w} \sum_{k \geq 0} C_j(k) \frac{w^k}{k!}. \quad (52)$$

This is a polynomial of degree at most j . Since $\mathcal{H}(0, \varepsilon) = 0$, it has $P_j(0) = 0$ for $j \geq 1$. This proves (36).

For the weighted assertion choose a circle $|z| = R'$ with $R' > \max(q, 1)$. The PGF error is bounded there by $M_{J,R'} \varepsilon^{J+1}$, so its ℓ th coefficient has modulus at most $M_{J,R'} \varepsilon^{J+1} R'^{-\ell}$. Summing with weight q^ℓ gives the factor $(1 - q/R')^{-1}$. Finally,

$$Q_j(\ell) = \sum_{h=0}^j [w^h] P_j(w) \sum_{b=0}^h \binom{h}{b} (-1)^{h-b} (\ell)_b, \quad (53)$$

where $(\ell)_b$ is the falling factorial. This proves its polynomial degree bound and the coefficient formula. $\square$

The $J = 0$ case recovers Poisson convergence with an $O(n^{-1/2})$ total-variation error, but the limit itself is prior work. Higher orders refine every fixed exponential moment and every polynomially weighted probability sum. The constants are for fixed weights; a weight $q = q_n \rightarrow \infty$ is outside the theorem.

Why the total variation has its own expansion

Proof of Corollary 9.2. The avoidance formula is (36) at $w = -1$. For total variation, first fix $J \geq 3$ and put

$$\mathcal{D}_J(\ell, \varepsilon) = \frac{e^{-1}}{\ell!} \sum_{j=1}^J Q_j(\ell) \varepsilon^j.$$

The weighted theorem with $q = 1$ changes the half- ℓ^1 norm by $O_J(\varepsilon^{J+1})$. At $\ell = 0$ the leading term of $\mathcal{D}_J$ is positive and of order ε . At $\ell = 1$, $Q_1(1) = Q_2(1) = 0$ and $Q_3(1) = 2/3 > 0$. For $\ell \geq 2$ the leading term is $-(\ell - 1)\varepsilon$ times $e^{-1}/\ell!$.

Take $\ell_0 = \lfloor (J+2) \log(1/\varepsilon) / \log \log(1/\varepsilon) \rfloor$. For $2 \leq \ell \leq \ell_0$,

$$\frac{\left| \sum_{j=2}^J \varepsilon^j Q_j(\ell) \right|}{(\ell - 1)\varepsilon} \leq C_J \sum_{j=2}^J [\varepsilon(1 + \ell_0)]^{j-1} = o(1).$$

The signs of this finite approximation are therefore positive at 0, 1 and negative on $2, \dots, \ell_0$. Its tail beyond ℓ_0 is bounded by $C_J \varepsilon \sum_{\ell > \ell_0} (1 + \ell)^J / \ell! = o(\varepsilon^{J+1})$. Each correction has mass zero. The half- ℓ^1 norm is consequently, to the stated error, the sum of the deviations at 0 and 1. These two coefficients are $e^{-1} P_j(-1)$ and $e^{-1} (P_j(-1) + P'_j(-1))$. This gives (40); lower orders follow by truncating the result for $J = 3$. $\square$

This proof needs no assertion about the exact deviation signs at every ℓ . Such an assertion would be stronger than the result established here.

9.7 Avoidance counts and their logarithms

Let $\kappa_j = P_j(-1)$ and define, coefficientwise,

$$d_j = \sum_{h=0}^j A_h \kappa_{j-h}, \quad \beta_j = [\varepsilon^j] \log \left(\sum_{h \geq 0} d_h \varepsilon^h \right).$$

Then for every fixed J ,

$$a_n = \frac{1}{\sqrt{2}} \left(\frac{n}{e} \right)^{n/2} e^{\sqrt{n}-5/4} \left(\sum_{j=0}^J d_j n^{-j/2} + O_J(n^{-(J+1)/2}) \right), \quad (54)$$

$$\log a_n = L_{1,J}(n) + O_J(n^{-(J+1)/2}), \quad (55)$$

$$L_{1,J}(x) = \frac{x}{2}(\log x - 1) + \sqrt{x} + c_A + \sum_{j=1}^J \beta_j x^{-j/2}, \quad c_A = -\frac{\log 2}{2} - \frac{5}{4}. \quad (56)$$

The first absolute and logarithmic coefficients are

$$(d_0, d_1, d_2, d_3, d_4) = \left(1, \frac{31}{24}, -\frac{359}{1152}, -\frac{93973}{414720}, \frac{15358133}{39813120} \right),$$

$$(\beta_1, \beta_2, \beta_3, \beta_4) = \left(\frac{31}{24}, -\frac{55}{48}, \frac{1717}{1920}, -\frac{75}{128} \right).$$

The d_1 term agrees with Kotěšovec's existing OEIS formula. The derivation also supplies each fixed subsequent order with a remainder; the coefficient table and executable extraction are included for use without manual algebra. Equations (55)–(56) are (27)–(28) at $r = 1$, with $\beta_j = \gamma_{1,j}$.

9.8 An exact decomposition into two Gaussian sectors

The algebraic expansion suppresses a parity effect exponentially small relative to the leading scale. An exact integral decomposition identifies it without assigning unjustified precision to a finite algebraic truncation.

Let $\mathcal{U}_n$ be the Chebyshev polynomial of the second kind. Its finite polynomial identity gives

$$\mathcal{U}_n(y/2) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} y^{n-2k},$$

that is, $\mathcal{U}_n(y/2) = \mu_{P_n}(y)$. Combining this with (43) at $\ell = 0$ and the Gaussian moment identity yields exactly

$$a_n = \int_{\mathbb{R}} \mathcal{U}_n(y/2) \phi(y-1) dy, \quad (57)$$

the case $G = P_n$ of Lemma 2.1. For $\sigma \in \{+1, -1\}$ define the positive integrals $\mathcal{K}_n^\sigma, \tau_n^\sigma$ and the

central term C_n^{cen} by

$$\begin{aligned}\mathcal{K}_n^\sigma &= \int_1^\infty \lambda^n \phi(\lambda + \lambda^{-1} - \sigma) d\lambda, \\ \tau_n^\sigma &= \int_1^\infty \lambda^{-n-2} \phi(\lambda + \lambda^{-1} - \sigma) d\lambda, \\ C_n^{\text{cen}} &= \int_{-2}^2 \mathcal{U}_n(y/2) \phi(y-1) dy.\end{aligned}$$

(Thus $C_n^{\text{cen}} = C_{P_n}$ and $K_{P_n}^\sigma = \mathcal{K}_n^\sigma - \tau_n^\sigma$ in the notation of (14).)

Theorem 9.5 (Exact sectors and separate expansions). *For every integer $n \geq 0$,*

$$a_n = \mathcal{K}_n^+ + (-1)^n \mathcal{K}_n^- + \Xi_n, \quad (58)$$

$$\Xi_n = C_n^{\text{cen}} - \tau_n^+ - (-1)^n \tau_n^-, \quad |\Xi_n| \leq n+1 + \frac{2}{\sqrt{2\pi}(n+1)}. \quad (59)$$

Put

$$\mathcal{L}_n^\sigma = \frac{1}{\sqrt{2}} \left(\frac{n}{e} \right)^{n/2} e^{\sigma\sqrt{n}-5/4} \quad (n \geq 1).$$

For each fixed $M \geq 0$ and each sign,

$$\mathcal{K}_n^\sigma = \mathcal{L}_n^\sigma \left(\sum_{m=0}^M \sigma^m d_m n^{-m/2} + O_M(n^{-(M+1)/2}) \right), \quad (60)$$

where d_m is exactly the coefficient in (54). In particular,

$$(-1)^n (a_n - \mathcal{K}_n^+) = \mathcal{L}_n^- \left(\sum_{m=0}^M (-1)^m d_m n^{-m/2} + O_M(n^{-(M+1)/2}) \right). \quad (61)$$

The constants can be chosen simultaneously for the two signs.

Exact decomposition. For $\lambda > 1$ use $y = \lambda + \lambda^{-1}$ and

$$\mathcal{U}_n(y/2) = \frac{\lambda^{n+1} - \lambda^{-n-1}}{\lambda - \lambda^{-1}}, \quad dy = (\lambda - \lambda^{-1}) \frac{d\lambda}{\lambda}.$$

Their product is $(\lambda^n - \lambda^{-n-2}) d\lambda$. Reflection on the negative exterior uses $\mathcal{U}_n(-y/2) = (-1)^n \mathcal{U}_n(y/2)$. Splitting (57) into the two exteriors and the central interval gives (58). On $[-1, 1]$, $|\mathcal{U}_n| \leq n+1$, so $|C_n^{\text{cen}}| \leq n+1$. Each τ_n^σ is at most $(\sqrt{2\pi}(n+1))^{-1}$, proving (59). $\square$

The central bound $n+1$ is sharper, for the path, than the bound $B^n = 2^n$ of (15); both are smaller than either sector scale by every inverse power of n .

Coefficient generator and remainder proof

The exact identity

$$-\frac{(\lambda + \lambda^{-1} - \sigma)^2}{2} = -\frac{(\lambda - \sigma)^2}{2} - 1 + \frac{\sigma}{\lambda} - \frac{1}{2\lambda^2}$$

shows how to expand each sector. With $\lambda = \sqrt{n} + u$, its normalized local integral is a Gaussian of mean $\sigma/2$ and variance $1/2$, multiplied by $e^{\mathcal{F}_\sigma(\varepsilon, u)}$, where

$$\mathcal{F}_\sigma(\varepsilon, u) = \frac{\log(1 + \varepsilon u) - \varepsilon u + (\varepsilon u)^2/2}{\varepsilon^2} + \frac{\sigma \varepsilon}{1 + \varepsilon u} - \frac{\varepsilon^2}{2(1 + \varepsilon u)^2}. \quad (62)$$

Define the coefficient polynomials

$$\begin{aligned} \hat{p}_{1,\sigma}(u) &= u^3/3 + \sigma, \\ \hat{p}_{h,\sigma}(u) &= (-1)^{h+1} \left(\frac{u^{h+2}}{h+2} + \sigma u^{h+1} + \frac{h-1}{2} u^{h-2} \right) \quad (h \geq 2), \\ \hat{b}_{0,\sigma}(u) &= 1, \quad \hat{b}_{m,\sigma}(u) = \frac{1}{m} \sum_{h=1}^m h \hat{p}_{h,\sigma}(u) \hat{b}_{m-h,\sigma}(u). \end{aligned}$$

For $V_\sigma \sim N(\sigma/2, 1/2)$ the sector coefficients are $\hat{D}_{m,\sigma} = \mathbb{E} \hat{b}_{m,\sigma}(V_\sigma)$. The symmetry $\hat{p}_{h,-}(-u) = (-1)^h \hat{p}_{h,+}(u)$ implies $\hat{D}_{m,-} = (-1)^m \hat{D}_{m,+}$.

On $|u| \leq \varepsilon^{-1/6}$, all denominators in (62) stay away from zero throughout the Taylor segment $0 \leq s \leq \varepsilon$. Its fixed derivatives have polynomial bounds in u , and $|\mathcal{F}_\sigma(s, u)| \leq C\varepsilon(1 + |u|^3) = O(\varepsilon^{1/2})$. Taylor's theorem gives an error at most $C_M \varepsilon^{M+1}(1 + |u|^{N_M})$, for some fixed N_M . Integration against the shifted Gaussian proves the local remainder.

For the tails set

$$f_{n,\sigma}(\lambda) = n \log \lambda - \frac{(\lambda - \sigma)^2}{2}, \quad g_\sigma(\lambda) = e^{-1 + \sigma/\lambda - 1/(2\lambda^2)}.$$

On $\lambda \geq 1$, $e^{-5/2} \leq g_\sigma(\lambda) \leq 1$. The maximum of $f_{n,\sigma}$ is at $y_{n,\sigma} = (\sigma + \sqrt{1 + 4n})/2 = \sqrt{n} + \sigma/2 + O(\varepsilon)$, and $f_{n,\sigma}'' \leq -1$. Its maximum value is $\frac{n}{2}(\log n - 1) + \sigma\sqrt{n} - 1/4 + O(\varepsilon)$. The omitted tails therefore contribute at most $\mathcal{L}_n^\sigma O(e^{-cn^{1/6}})$. This establishes (60) with $\hat{D}_{m,+}$ in place of d_m .

The negative sector and the polynomially bounded residual are smaller than the positive scale by every inverse power. Uniqueness of finite-order Poincaré coefficients in (54) and (58) gives $\hat{D}_{m,+} = d_m$ at every order. Finally, $\Xi_n/\mathcal{L}_n^-$ is also smaller than every inverse power, which proves (61).

[Added 2 October 2026, batch 77.] Theorem 3.3 at $G = P_n$ gives the same expansion for the sectors $K_{P_n}^\sigma = \mathcal{K}_n^\sigma - \tau_n^\sigma$ with the scale $L_n^\sigma = e^{1/n} \mathcal{L}_n^\sigma$ and the moving statistics $a_j(P_n)$; re-expanding $e^{1/n}$ and the affine a_j (Theorem 5.1) turns $D_h(a(P_n))$ into d_h (for instance $D_1 = 7/24 + a_1 \rightarrow 31/24 = d_1$). Source 26's generator $\hat{p}_{h,\sigma}$ already has these constants built in, which is why it differs from (19).

The meaning of the exponentially small term

The ratio of exact sectors satisfies

$$\frac{\mathcal{K}_n^-}{\mathcal{K}_n^+} = e^{-2\sqrt{n}}(1 + O(n^{-1/2})).$$

More concretely, with the exact positive-sector integral in the denominator,

$$\frac{a_n - \mathcal{K}_n^+}{\mathcal{K}_n^+} = (-1)^n e^{-2\sqrt{n}} \left(1 - \frac{31}{12\sqrt{n}} + O(n^{-1}) \right).$$

Thus $\mathcal{K}_n^+$ lies below a_n for all sufficiently large even n , and above it for all sufficiently large odd n . The small-sector statement (61) subtracts the *exact* $\mathcal{K}_n^+$. For a fixed M -term approximation to that integral, the available error bound is $O(\mathcal{L}_n^+ n^{-(M+1)/2})$, whose scale is much larger than $\mathcal{L}_n^-$. Thus a finite dominant-sector truncation does not numerically or analytically resolve the smaller term.

Equivalently, one may write a joint formula with two independent errors:

$$\begin{aligned} a_n &= \mathcal{L}_n^+ \left(\sum_{m=0}^M d_m \varepsilon^m + E_{n,M}^+ \right) \\ &\quad + (-1)^n \mathcal{L}_n^- \left(\sum_{m=0}^N (-1)^m d_m \varepsilon^m + E_{n,N}^- \right) + \Xi_n, \\ |E_{n,M}^+| &\leq C_M \varepsilon^{M+1}, \quad |E_{n,N}^-| \leq C_N \varepsilon^{N+1}. \end{aligned}$$

This is an exact two-sector decomposition with separate Poincaré expansions. The word “transseries” may describe its formal two-scale structure, but no optimal-truncation, Borel-summability, resurgence, or Stokes-constant theorem is established here.

9.9 Inversion with three distinct conventions

The count grows superexponentially. Its inverse must distinguish a smooth asymptotic root, an integer threshold, and the inverse of an actual log-linear interpolation.

Smooth roots, the finite Newton algorithm and the integer threshold. With $L_{1,J}$ from (56), source 26 states and proves, in the same form, Section 6 at $r = 1$: the unique large root $x_{1,J}(y)$ of $L_{1,J}(x) = \log y$ with $x_{1,J}(a_n) = n + O_J(n^{-(J+1)/2}/\log n)$ (30); the Lambert- W seed and Newton iteration (31) with the error bound of Section 6, stated for every fixed iteration count $i \geq 0$ (the initial error has precisely the stated order at $i = 0$; the first update is much smaller than the initial neighborhood, so all iterates remain in the range where the bounds apply), and

$$i = \lceil \log_2(J+3) \rceil \tag{63}$$

updates; and, for $N_1(y) = \min\{n \geq 0 : a_n \geq y\}$, strict increase of a_n for $n \geq 3$ and the two-ceiling enclosure of Proposition 6.1 at $r = 1$, with the finite Newton iterate allowed in place of x . The ceiling cannot in general be discarded, even though $\delta \rightarrow 0$: a threshold may be arbitrarily close to one of the sequence values. The proof supplies existence of the constants; the package does not claim a ready-to-use numerical certification of C_J and y_J . The relation of these statements to the repository’s inversion theorems is the note at the end of Section 6.

The actual log-linear interpolant. Define $\mathcal{A}_1(x)$ for $x \geq 3$ by

$$\log \mathcal{A}_1(m + \theta) = (1 - \theta) \log a_m + \theta \log a_{m+1}, \quad m \in \mathbb{Z}, \quad 0 \leq \theta \leq 1.$$

Its inverse is an unambiguous continuous index convention. To approximate it at arbitrarily high order, interpolate the asymptotic logarithm at the same nodes:

$$\tilde{L}_{1,J}(m + \theta) = (1 - \theta)L_{1,J}(m) + \theta L_{1,J}(m + 1).$$

Let $\tilde{x}_{1,J}(y)$ be the large root of $\tilde{L}_{1,J}(x) = \log y$. On the interval whose endpoint values bracket $b = \log y$, it is explicitly

$$\tilde{x}_{1,J}(y) = m + \frac{b - L_{1,J}(m)}{L_{1,J}(m+1) - L_{1,J}(m)}. \tag{64}$$

Proposition 9.6 (Log-linear convention). *For fixed J , writing $x = \tilde{x}_{1,J}(y)$,*

$$\mathcal{A}_1^{-1}(y) = \tilde{x}_{1,J}(y) + O_J \left(\frac{x^{-(J+1)/2}}{\log x} \right).$$

In contrast, the difference between the smooth and piecewise-linear roots is generally only bounded by $O(1/(x \log x))$, independently of the truncation order.

Proof. The nodal error (55) interpolates linearly to a uniform $O_J(x^{-(J+1)/2})$ error on each unit interval. The slopes of $\tilde{L}_{1,J}$ are $\Theta(\log x)$, so inversion gives the first assertion. Linear interpolation of a twice differentiable function differs from that function by $O(\sup |L''_{1,J}|) = O(1/x)$ on a unit interval. Its inverse discrepancy is therefore $O(1/(x \log x))$.

More precisely, for $x = m + \theta$, Taylor's theorem gives

$$\tilde{L}_{1,J}(x) - L_{1,J}(x) = \frac{\theta(1-\theta)}{4x} + O_J(x^{-3/2}).$$

Away from the endpoints this leading chord term is nonzero; it cannot be ignored in an arbitrary-order continuous inverse claim. $\square$

Proposition 9.6 is (34) at $r = 1$, with the explicit leading chord term added. Finding the correct interval for (64) requires only checking nearby integer endpoint values of $L_{1,J}$, starting from the smooth Newton approximation. This involves evaluations of the explicit truncation, not a conjectural interpolation of a_n .

9.10 Computation and reproducibility of source 26

The accompanying ZIP contained the editable L^AT_EX source and PDF, four mathematical Python scripts and a comparison helper, exact rational coefficient data, numerical check outputs, a coefficient supplement, source provenance, content hashes, and a replay script. It contains no downloaded third-party papers or Li manuscript source.

[Added 2 October 2026, batch 77.] In this report the scripts are `code/26-adjacent-scripts-*.py`, the replay and build scripts `code/26-adjacent-replay.sh` and `code/26-adjacent-build.sh`, the data `data/26-adjacent-*`, the supplements `26-adjacent-supplements-*.md`; the delivered source `report.tex` is printed as this section, and the delivered PDF and the checksum list `SHA256SUMS` (which the delivered `replay.sh` verifies first) are not shipped.

The extraction script uses exact rational arithmetic. If $q_h(u) = (-1)^{h+3}u^{h+2}/(h+2)$ and $e^{\sum_{h \geq 1} q_h \varepsilon^h} = \sum_{m \geq 0} f_m(u) \varepsilon^m$, then

$$f_0 = 1, \quad f_m = \frac{1}{m} \sum_{h=1}^m h q_h f_{m-h}.$$

Gaussian expectation gives A_m . Formal logarithm, the explicit series $\mathcal{H}$, and a Touchard transform give all the other coefficient families. A second generator uses the exact sector integral and agrees through the checked orders; the proof establishes equality at all finite orders.

The supplied checks cover:

- exact triangle rows $n = 0, \dots, 8$ and the avoidance recurrence;

- avoidance-ratio residuals through $n = 20000$;
- total-variation and finite-Newton residuals at $n = 100, 1000, 5000, 20000$;
- independent sector coefficients through order 10;
- the exact sector identity at $n = 0, 1, 2, 3, 4, 8, 15, 20$, and separate normalized sector residuals at $n = 100, 400, 1600$.

These are reproducibility checks, not replacements for the analytic proofs. The total-variation numerical routine truncates the moment series at $K \leq 90$; its ℓ^1 tail is bounded by $\sum_{k>K} 2^k/k!$, and the omitted Poisson mass must also be included when interpreting precision. At $K = 90$ the bound is tiny, but the displayed decimal strings are not claimed as fully certified interval enclosures.

n	$\varepsilon^{-3}(ed_{\text{TV}} - \varepsilon + \varepsilon^2/2)$	inverse $J = 3$ scaled error	inverse $J = 7$ scaled error
100	0.6686654	-1.1769375	8.5338127
1000	0.6944681	-1.1799798	9.1775393
5000	0.7019778	-1.1765105	9.4307369
20000	0.7051241	-1.1744492	9.5334214

The last two columns report $(x_i - n)n^{(J+1)/2} \log n$ at $y = a_n$, using the iteration count (63). Their limits are $2\beta_4 = -75/64$ and $2\beta_8 = 19733/2048$, respectively. The first column tends to $17/24$. No monotonic approach is required by the error theorems. (The table is `data/26-adjacent-numerical-table.tex`.)

9.11 Further research questions of source 26

The following are directions suggested by the proofs, not results proved in source 26.

1. **Effective constants.** Can one obtain modest explicit values for the uniform PGF, weighted-law, and inverse-enclosure constants, so that a requested finite n tolerance is numerically certified?
2. **Growing parameters.** How far may the complex PGF argument, the exponential weight, or the deviation index grow with n while retaining a useful uniform expansion? The present fixed-compact theorem does not answer a moderate- or large-deviation question.
3. **Other sparse forbidden graphs.** Replace the path by a bounded-degree graph on the labels. Its matching counts replace $\binom{n-k}{k}$. Which graph families admit polynomial coefficient structures and weighted Poisson refinements with comparably simple errors?

[Added 2 October 2026, batch 77.] *Re-scoped.* Source 19 answers the weighted-Poisson part for *every* family of bounded maximum degree: Theorem 3.1 and Corollary 3.2 give the complex-PGF expansion and the weighted signed-law error to every fixed order, with universal rational polynomial coefficients in the graph's actual root moments $a_j(G)$, and Theorem 3.3 the two sectors; Theorem 5.1 gives constant coefficients for every fixed finite forbidden-distance set. What remains open is which other families have convergent or eventually affine statistics $a_j(G_n)$ (so that the coefficients become constant), and the efficiency questions of Section 8.

4. **Small-sector resolution.** What is the large-order growth of d_j , and can optimal truncation of the positive sector reach the $e^{-2\sqrt{n}}$ scale? A proof would require tools beyond the separate fixed-order Laplace estimates used here.
5. **Joint laws.** Can the adjacent-transposition count be refined jointly with the total number of transpositions, fixed points, or crossings, and what covariance corrections survive in natural scalings?
6. **Historically complete comparison.** A broader inspection of the computational-biology and involution asymptotics literature may locate overlapping refinements. Establishing priority is a separate bibliographic task.

These questions preserve the useful elementary core: exact matching enumeration, Gaussian asymptotics, and a remainder that survives summation over the whole distribution.

A Selected coefficients of the adjacent case through order eight

This appendix is source 26's. Set $\kappa_j = P_j(-1)$. The exact rational tables below use the normalizations in (39), (40), and (55). The complete C_j and P_j are in the coefficient supplement `26-adjacent-supplements-coefficients.md` and the JSON data.

j	κ_j	V_j	β_j
0	1	0	0
1	1	1	31/24
2	-1/2	-1/2	-55/48
3	1/24	17/24	1717/1920
4	7/24	-11/24	-75/128
5	-1759/1920	1217/1920	-26879/107520
6	1487/1440	-233/1440	899/720
7	-482159/322560	-93119/322560	-6385907/2064384
8	8591/10080	11551/5040	19733/4096

All displayed decimals in the computational outputs derive from exact fractions or high-precision quadrature. The analytic results concern exact quantities.

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